

# ANOVA & MANOVA

## Complete Exam Answer Guide

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ANOVA Assumptions

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MANOVA Stats

Based strictly on: 18.-Hypothesis-Testing.pdf (Tushar B. Kute) and Chapter 10 — One Sample Tests of Hypothesis (Course PPT)

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**Illustrate the assumptions of ANOVA. (Independence, Normality, Homogeneity of Variance, Random Sampling)**

## What is ANOVA?

ANOVA (Analysis of Variance) is a statistical test used to analyse the difference between the means of more than two groups simultaneously. It tests whether the dependent variable changes according to the level of the independent variable(s). ANOVA uses the F-test for statistical significance — comparing variance between groups to variance within groups.

$H_0: \mu_1 = \mu_2 = \dots = \mu_k$  (All group means are equal)

$H_1$ : At least one group mean is DIFFERENT (Not all means are equal)

## 4 Assumptions of ANOVA — Detailed Explanation

From your notes: 'The assumptions of the ANOVA test are the same as the general assumptions for any parametric test.' There are 4 core assumptions:

1

### Independence of Observations

The data must have been collected using statistically valid methods and there must be NO hidden relationships among the observations. Each subject/unit must belong to exactly ONE group and must be measured independently of all other subjects. From your notes: 'If your data fail to meet this assumption because you have a confounding variable that you need to control for statistically, use an ANOVA with blocking variables.' Example violation: If students in the same classroom are assigned to different treatment groups, they may influence each other — violating independence.

2

### Normality of the Response Variable

The values of the dependent variable must follow a NORMAL DISTRIBUTION within each group being compared. From your notes: 'The values of the dependent variable follow a normal distribution.' ANOVA is considered fairly ROBUST to mild departures from normality, especially with large sample sizes (Central Limit Theorem applies). Detection: Q-Q plot of residuals per group; Shapiro-Wilk test on each group's data. Example: Exam scores in each teaching-method group should be approximately bell-shaped.

**3****Homogeneity of Variance (Homoscedasticity)**

The variance within each group being compared must be APPROXIMATELY EQUAL across all groups. This is sometimes called the equal-variance assumption. From your notes: 'The variation within each group being compared is similar for every group. If the variances are different among the groups, then ANOVA probably is not the right fit for the data.' Detection: Levene's test; Bartlett's test; comparing the ratio of the largest to smallest group variance (should be less than 4:1 as a rough rule). Example violation: If one fertilizer group has very high variance in crop yield while another group has very low variance, the assumption is violated.

**4****Random Sampling**

Data must be collected through RANDOM SAMPLING so that each unit in the population has an equal chance of being selected into the study. This ensures the sample is representative of the population and that results can be generalised. Example: In a drug study comparing 3 medications, patients must be randomly assigned to one of the three treatment groups — not self-selected.

## Why Are These Assumptions Important?

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When the assumptions are met, the F-test used in ANOVA is valid and the results can be trusted. Violating assumptions leads to incorrect p-values, wrong conclusions about group differences, and unreliable inference.

What to do when assumptions are violated: Independence violated: Redesign the study. Use mixed-effects models. Normality violated: Use Kruskal-Wallis test (non-parametric alternative to one-way ANOVA) or apply a data transformation (log, square root). Homogeneity of variance violated: Use Welch's ANOVA (does not assume equal variances) or apply data transformation. Random sampling violated: Results cannot be generalised beyond the specific sample.

**Explain the use of Two-way ANOVA with a suitable example. (Two factors, interaction term, F-statistic for each factor)**

## What is Two-way ANOVA?

Two-way ANOVA extends one-way ANOVA by examining the effect of TWO independent variables (factors) on a single dependent variable simultaneously. It also tests whether there is an INTERACTION between the two factors — meaning whether the effect of one factor depends on the level of the other factor.

One-way ANOVA: One factor (e.g. teaching method) on one dependent variable (exam score)  
Two-way ANOVA: Two factors (e.g. teaching method AND gender) on one dependent variable (exam score)  
Also tests: Does the effect of teaching method DIFFER between males and females? (This is the interaction effect)

## What Two-way ANOVA Tests — Three Hypotheses

Two-way ANOVA tests THREE separate hypotheses simultaneously:

| Test                  | H0 (Null)                                                   | H1 (Alternate)                                            | F-statistic           |
|-----------------------|-------------------------------------------------------------|-----------------------------------------------------------|-----------------------|
| Main effect: Factor A | Factor A has NO effect on Y<br>(all row means are equal)    | Factor A DOES affect Y                                    | $F_A = MSA / MSE$     |
| Main effect: Factor B | Factor B has NO effect on Y<br>(all column means are equal) | Factor B DOES affect Y                                    | $F_B = MSB / MSE$     |
| Interaction: A x B    | No interaction between Factor A and Factor B                | The effect of A depends on the level of B (or vice versa) | $F_{AB} = MSAB / MSE$ |

## SS Decomposition in Two-way ANOVA

The total variation is decomposed into four parts:

$$SST (Total) = \sum \sum (Y_{ij} - Y_{grand\_bar})^2$$

*Total variation in Y around the grand mean*

|                    |                                                                                                             |
|--------------------|-------------------------------------------------------------------------------------------------------------|
| SSA (Factor A)     | $= n_B * \sum (Y_{A\_bar\_i} - Y_{grand\_bar})^2$<br><i>Variation due to Factor A (between levels of A)</i> |
| SSB (Factor B)     | $= n_A * \sum (Y_{B\_bar\_j} - Y_{grand\_bar})^2$<br><i>Variation due to Factor B (between levels of B)</i> |
| SSAB (Interaction) | $= SST - SSA - SSB - SSE$<br><i>Variation due to the A-by-B interaction</i>                                 |
| SSE (Error)        | $= \sum \sum (Y_{ij} - \hat{Y}_{ij})^2$<br><i>Residual variation within cells</i>                           |
| Decomposition      | $SST = SSA + SSB + SSAB + SSE$<br><i>Total = Factor A + Factor B + Interaction + Error</i>                  |

$$F_A = MSA / MSE \mid F_B = MSB / MSE \mid F_{AB} = MSAB / MSE$$

$$MS = SS / df \mid df_A = a-1 \mid df_B = b-1 \mid df_{AB} = (a-1)(b-1) \mid df_E = ab(n-1)$$

Decision rule: Reject  $H_0$  if  $F_{calculated} > F_{critical}$  at significance level  $\alpha$ . Equivalently: Reject  $H_0$  if  $p\text{-value} < \alpha$ . IMPORTANT: Always check the interaction effect FIRST. If the interaction is significant, the main effects cannot be interpreted in isolation — the effect of one factor depends on the level of the other.

## Full Worked Example — Crop Yield Study

Research question: Does crop yield depend on (A) Fertilizer type (3 types: F1, F2, F3) and (B) Irrigation method (2 types: Drip, Flood)? Is there an interaction between fertilizer and irrigation?

### Setting up the three sets of hypotheses:

|                       |                                                                                                                                                        |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Factor A (Fertilizer) | $H_0: \mu_{F1} = \mu_{F2} = \mu_{F3}$ (fertilizer type has no effect on yield)<br>$H_1$ : At least one fertilizer gives different mean yield           |
| Factor B (Irrigation) | $H_0: \mu_{Drip} = \mu_{Flood}$ (irrigation method has no effect on yield)<br>$H_1$ : Drip and Flood irrigation give different mean yields             |
| Interaction A x B     | $H_0$ : The effect of fertilizer is the SAME for both irrigation methods<br>$H_1$ : The effect of fertilizer DIFFERS between drip and flood irrigation |

### Interpreting results:

If  $F_{AB}$  is significant: the best fertilizer for drip irrigation may differ from the best fertilizer for flood irrigation. Report and interpret the interaction. If  $F_{AB}$  is NOT significant: interpret  $F_A$  and  $F_B$  independently. If  $F_A$  is significant, use post-hoc tests (Tukey's HSD) to find which fertilizer pairs differ.

When to use Two-way ANOVA: • You have ONE quantitative dependent variable • You have exactly TWO categorical independent variables (factors) • Each factor has at least 2 levels • You want to know: (1) effect of A, (2) effect of B, (3) whether A and B interact • If you have 3 or more factors, use factorial ANOVA (multi-way ANOVA)

**Explain the use of One-way MANOVA with a suitable example. (Multiple dependent variables, Pillai's trace, Wilks' lambda)**

## What is MANOVA?

MANOVA (Multivariate Analysis of Variance) is an extension of ANOVA that handles MULTIPLE DEPENDENT VARIABLES simultaneously. While ANOVA tests whether group means differ on a single dependent variable, MANOVA tests whether group means differ across a combination of dependent variables at the same time.

### ANOVA

1 independent variable (factor) 1 dependent variable  
method affect exam scores?

### MANOVA

1 independent variable (factor) 2+ dependent variables (outcome variables)  
method affect BOTH exam scores AND attendance?

## One-way MANOVA — Hypotheses

In one-way MANOVA, there is ONE factor (independent variable) and TWO OR MORE dependent variables (outcome variables).

H0: The mean VECTOR of all dependent variables is equal for all groups.  
 $\mu_{1\_Y2} = \mu_{2\_Y1}, \mu_{2\_Y2} = \dots = \mu_{k\_Y1}$

H1: At least one group has a different mean VECTOR of dependent variables.  
group differs on at least one dependent variable

## MANOVA Test Statistics

Unlike ANOVA which uses a single F-statistic, MANOVA uses multivariate test statistics that account for all dependent variables simultaneously and their correlations with each other. The four main statistics are:

| Statistic      | How It Works                                                                                                                           | Interpretation                                                                                                        |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Pillai's Trace | Sum of proportions of variance in the dependent variables explained by the factor. Ranges from 0 to the number of dependent variables. | Larger value = stronger group separation. Most ROBUST statistic — recommended when assumptions are not perfectly met. |



|                    |                                                                                                    |                                                                                                                                    |
|--------------------|----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Wilks' Lambda      | Ratio of within-group variance to total variance. Ranges from 0 to 1.                              | Closer to 0 = strong group separation (large effect). Closer to 1 = weak or no group separation. Most commonly reported statistic. |
| Hotelling's Trace  | Sum of eigenvalues of the between-groups variance matrix divided by within-groups variance matrix. | Larger value = stronger group separation. Suitable when there are only 2 groups.                                                   |
| Roy's Largest Root | Largest eigenvalue of the between-to-within variance ratio matrix.                                 | Most powerful when there is a single strong discriminating dimension between groups. Sensitive to violations of assumptions.       |

Which statistic to report? Pillai's Trace is the most robust and is recommended when sample sizes are unequal or when assumptions may be slightly violated. Wilks' Lambda is the most widely reported in published research.

## Assumptions of MANOVA

1

### Independence of Observations

Each participant/unit belongs to exactly one group and is measured independently.

2

### Multivariate Normality

The dependent variables are jointly normally distributed within each group. MANOVA is robust to mild violations especially with large samples.

3

### Homogeneity of Variance-Covariance Matrices

The variance-covariance matrix of the dependent variables must be equal across groups. Tested with Box's M test. MANOVA is somewhat robust to this violation.

4

### No Multicollinearity Between DVs

The dependent variables should be correlated with each other (that is why we use MANOVA instead of separate ANOVAs), but not perfectly correlated (which would be redundant).

5

### No Outliers

Multivariate outliers can heavily distort MANOVA results. Check using Mahalanobis distance.

## Full Worked Example — Teaching Methods Study

Research question: Do three different teaching methods (Traditional, Flipped Classroom, Online) produce different outcomes on students? Independent variable (Factor): Teaching method (3 groups / levels) Dependent variables: (1) Final exam score and (2) Course attendance percentage Why use MANOVA instead of two separate ANOVAs? Because exam score and attendance are correlated — students who attend more also tend to score higher. Running two separate ANOVAs ignores this correlation and inflates the Type I error rate.

### Steps:

#### 1 State hypotheses

H0:  $[\mu_{\text{exam}}, \mu_{\text{attend}}]$  is equal across all 3 teaching methods. H1: At least one method has a different mean vector.

#### 2 Check assumptions

Verify independence, multivariate normality (Q-Q plots per group), homogeneity of covariance matrices (Box's M test).

#### 3 Run MANOVA and check test statistics

Report Pillai's Trace and Wilks' Lambda. If  $p < 0.05$  for any statistic, reject H0 — at least one group differs on the combination of exam score and attendance.

#### 4 Run follow-up ANOVAs (if MANOVA significant)

Run separate one-way ANOVAs for each DV to identify WHICH dependent variable shows the group difference. Apply Bonferroni correction ( $\alpha$  / number of DVs) to control Type I error.

#### 5 Post-hoc tests

If ANOVA is significant, run Tukey's HSD or Bonferroni post-hoc tests to identify WHICH specific group pairs differ.

MANOVA result interpretation example: Wilks' Lambda = 0.42,  $F(4, 116) = 8.34$ ,  $p < 0.001$ . Since  $p < 0.05$ , reject H0. The three teaching methods produce significantly different outcomes on the combined dependent variable (exam score + attendance). Follow-up ANOVA on exam score:  $F(2,59) = 9.21$ ,  $p < 0.001$  — significant Follow-up ANOVA on attendance:  $F(2,59) = 4.18$ ,  $p = 0.02$  — significant Both dependent variables show group differences.

Explain ANOVA with a suitable example and all its assumptions. (Full treatment:  $H_0$ , F-test, SSB, SSW, decision rule)

## Definition and Purpose

ANOVA (Analysis of Variance) is a statistical method used to test whether there are statistically significant differences between the means of THREE OR MORE groups. It uses the F-test and compares the variation BETWEEN groups to variation WITHIN groups. From your notes: 'ANOVA determines whether the groups created by the levels of the independent variable are statistically different by calculating whether the means of the treatment levels are different from the overall mean of the dependent variable.'

Why not just use multiple t-tests? If we compare 4 groups using pairwise t-tests at  $\alpha=0.05$ , we need  $C(4,2)=6$  tests. Overall Type I error =  $1 - (0.95)^6 = 26.5\%$ . ANOVA tests all groups at once and controls the error rate at exactly  $\alpha$ .

## Hypotheses

$H_0: \mu_1 = \mu_2 = \mu_3 = \dots = \mu_k$  (ALL  $\mu$  are equal, independent variable has NO effect)

$H_1$ : At least one  $\mu_i$  is DIFFERENT from the others (At least one independent variable HAS an effect)

## The F-Statistic — SS Decomposition

ANOVA decomposes the Total Sum of Squares (SST) into two components:

### SSB — Between Groups

Measures variation between group means and the grand mean. This is the variation EXPLAINED by the grouping factor.

### SSW — Within Groups

Measures variation within each group — the random variation that CANNOT be explained by the grouping factor (error/residual).

$$SSB = \sum_i n_i (x_{\cdot i} - x_{\cdot\cdot})^2$$

$n_i$  = size of group  $i$   $x_{\cdot i}$  = mean of group  $i$

$$SSW = \sum_i \sum_j (x_{ij} - x_{\cdot i})^2$$

$x_{ij}$  = each observation in group  $i$   $x_{\cdot i}$  = mean of group  $i$

$$MSB = SSB / (k - 1)$$

Mean Square Between |  $df_{\text{between}} = k - 1$

$$MSW = SSW / (N - k)$$

Mean Square Within |  $df_{\text{within}} = N - k$  |  $N$  = total observations

$$SST = SSB + SSW$$

Total variation = Between-group variation + Within-group variation

$$F = MSB / MSW = [SSB / (k - 1)] / [SSW / (N - k)]$$

A large  $F$  means between-group variance  $\gg$  within-group variance — strong evidence that group means differ

## Decision Rule

Reject  $H_0$  if:  $F_{\text{calculated}} > F_{\text{critical}}(k-1, N-k, \alpha)$  The  $F_{\text{critical}}$  value is found from the F-distribution table with: Numerator degrees of freedom =  $k - 1$  (between groups)  
Denominator degrees of freedom =  $N - k$  (within groups) Alternatively: Reject  $H_0$  if  $p\text{-value} < \alpha$   
If  $H_0$  is rejected: use POST-HOC tests (Tukey's HSD, Bonferroni) to identify WHICH specific group pairs have significantly different means.

## 4 Assumptions of ANOVA

1

### Independence of Observations

All observations are independent. Data collected using statistically valid random sampling. No hidden relationships among observations.

## 2 Normality of Response Variable

The dependent variable follows a normal distribution within each group.  
Robust to mild departures with large samples (CLT applies).

## 3 Homogeneity of Variance

The variance within each group is approximately equal across all groups.  
Tested with Levene's test or Bartlett's test.

## 4 Random Sampling

Participants must be randomly assigned or selected into groups to ensure results generalise to the population.

## Full Worked Example — Medication Treatments

From your notes: 'We can use ANOVA to prove/disprove if all the medication treatments were equally effective or not.' Scenario: A pharmaceutical researcher tests whether 3 blood pressure medications (Drug A, Drug B, Drug C) produce different mean reductions in blood pressure. 15 patients are randomly assigned — 5 per drug. Blood pressure reduction (mm Hg) measured.

| Group      | n <sub>i</sub> | Mean (x-bar <sub>i</sub> ) | Interpretation                          |
|------------|----------------|----------------------------|-----------------------------------------|
| Drug A     | 5              | 12 mm Hg                   | Moderate reduction                      |
| Drug B     | 5              | 18 mm Hg                   | Higher reduction                        |
| Drug C     | 5              | 10 mm Hg                   | Lowest reduction                        |
| Grand Mean | 15             | 13.33 mm Hg                | $\bar{x}_{\text{grand}} = (12+18+10)/3$ |

### Calculating SS:

*H0 and H1*

$H_0: \mu_A = \mu_B = \mu_C$  |  $H_1$ : At least one differs  
*k=3 groups, N=15 total*

*SSB*

$= 5*(12-13.33)^2 + 5*(18-13.33)^2 + 5*(10-13.33)^2 = 5*1.77 + 5*21.77 + 5*11.09 = 173.33$   
*Between-group sum of squares*

*SSW*

= sum of squared deviations within each drug group  
*Within-group sum of squares (from raw data)*

*MSB*

$= SSB / (k-1) = 173.33 / 2 = 86.67$   
*df<sub>between</sub> = k-1 = 3-1 = 2*

|                 |                                                                                                                                                                                            |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>MSW</i>      | $= SSW / (N-k) = SSW / 12$<br>$df_{within} = N-k = 15-3 = 12$                                                                                                                              |
| <i>F</i>        | $= MSB / MSW$<br><i>Compare to <math>F_{critical}(2, 12, 0.05) = 3.89</math></i>                                                                                                           |
| <i>Decision</i> | <p>If <math>F &gt; 3.89</math>: Reject <math>H_0</math> and conclude the drugs differ in effectiveness</p> <p><i>Then run Tukey's HSD post-hoc to find which specific pairs differ</i></p> |

**Explain the difference between one-way ANOVA and two-way ANOVA. When would you choose MANOVA over ANOVA?**

### One-way ANOVA vs Two-way ANOVA — Full Comparison

| Aspect             | One-way ANOVA                                    | Two-way ANOVA                                                                        |
|--------------------|--------------------------------------------------|--------------------------------------------------------------------------------------|
| Number of factors  | ONE independent variable (factor)                | TWO independent variables (factors)                                                  |
| Number of DVs      | ONE dependent variable                           | ONE dependent variable                                                               |
| Hypotheses tested  | 1 hypothesis: Do group means on Factor A differ? | 3 hypotheses: (1) Main effect of A (2) Main effect of B (3) Interaction A*B          |
| Interaction effect | Cannot test — only one factor present            | YES — tests if effect of A depends on level of B                                     |
| F-statistics       | 1 F-statistic: $F = MSB / MSW$                   | 3 F-statistics: $F_A$ , $F_B$ , $F_{AB}$                                             |
| Example            | Do 3 fertilizers differ in crop yield?           | Do fertilizer type AND irrigation method affect crop yield? Is there an interaction? |
| When to use        | One categorical IV, one numeric DV               | Two categorical IVs, one numeric DV, especially when interaction matters             |

### When to Choose MANOVA over ANOVA

MANOVA should be chosen over ANOVA when you have MULTIPLE DEPENDENT VARIABLES that are correlated with each other. The decision guide:

| Situation                                     | Correct test                             |
|-----------------------------------------------|------------------------------------------|
| 1 factor, 1 dependent variable                | <b>One-way ANOVA</b>                     |
| 2 factors, 1 dependent variable               | <b>Two-way ANOVA</b>                     |
| 1 factor, 2+ dependent variables (correlated) | <b>One-way MANOVA</b>                    |
| 2 factors, 2+ dependent variables             | <b>Two-way MANOVA (factorial MANOVA)</b> |

Compare only 2 group means, 1 DV

**t-test (not ANOVA)**

## 5 Reasons to Choose MANOVA Over Separate ANOVAs

### 1 Controls Type I Error Rate

Running multiple separate ANOVAs on correlated DVs inflates the overall Type I error rate (false positives accumulate). MANOVA tests all DVs simultaneously and maintains the error rate at alpha.

### 2 Accounts for Correlations Between DVs

MANOVA exploits the correlations among dependent variables. If DVs are highly correlated, MANOVA is more powerful than running separate ANOVAs.

### 3 Detects Multivariate Effects

A group difference may only exist on the COMBINATION of DVs — not on any single DV alone. MANOVA can detect these multivariate effects that separate ANOVAs would miss.

### 4 Reduces Redundancy

When DVs measure related constructs (e.g. different subscales of the same test), MANOVA avoids the redundancy of multiple separate tests.

### 5 Single Omnibus Test

MANOVA provides a single overall test of the hypothesis, making interpretation cleaner and more coherent than interpreting many separate ANOVA results.

When NOT to use MANOVA: • When dependent variables are completely UNRELATED — there is no benefit • When dependent variables are perfectly correlated — causes multicollinearity • When sample sizes are very small relative to the number of DVs • When the assumptions (especially multivariate normality) are severely violated Rule of thumb: Use MANOVA when DVs are moderately correlated ( $r = 0.3$  to  $0.7$ ). If DVs are uncorrelated ( $r$  near 0), separate ANOVAs work just as well.

## Quick Reference Summary Table

| Test          | IVs           | DVs       | Key statistic | Tests                  |
|---------------|---------------|-----------|---------------|------------------------|
| One-way ANOVA | 1 categorical | 1 numeric | $F = MSB/MSW$ | Group mean differences |



|                |               |            |                              |                                           |
|----------------|---------------|------------|------------------------------|-------------------------------------------|
| Two-way ANOVA  | 2 categorical | 1 numeric  | F_A, F_B, F_AB               | Main effects + interaction                |
| One-way MANOVA | 1 categorical | 2+ numeric | Wilks Lambda, Pillai's Trace | Multivariate group differences            |
| Two-way MANOVA | 2 categorical | 2+ numeric | Wilks Lambda, Pillai's Trace | Main effects + interaction (multivariate) |